

XNBR-RELATED PATENT LANDSCAPE: ELASTOMER NANOCOMPOSITES AND SWELLABLE DOWNHOLE SEAL SYSTEMS

1. ABSTRACT

This report reviews two primary US patents that are relevant to XNBR (carboxylated nitrile butadiene rubber) through adjacent elastomer-processing and downhole sealing technologies rather than direct XNBR polymerization routes. US20120035309A1 discloses a method for dispersing nanoparticles into elastomers by first forming a nanoparticle pre-blend in a liquid form additive such as oil, plasticizer, or solvent, then blending that pre-blend with an elastomer to create a homogeneous nanocomposite. The patent explicitly includes XNBR among the elastomers that can be used, and emphasizes improved nanoparticle dispersion, reduced agglomeration, and suitability for harsh downhole environments. US8490707B2 discloses swellable sealing systems for subterranean wellbores using a base polymer combined with reactive fillers, including metal oxides, to produce a seal assembly that swells and stiffens in response to a triggering fluid and may remain swollen after fluid removal. Although this patent is not a synthesis patent for XNBR, it is highly relevant to XNBR application development because it addresses the same oilfield sealing use case and highlights filler-driven reinforcement strategies that can be applied to nitrile-based elastomers. Across the set, the dominant formulation/process themes are pre-dispersion of functional additives, use of liquid carriers or reactive fillers to improve dispersion and mechanical performance, and design for high-temperature/high-pressure downhole service. No direct XNBR polymerization recipe is disclosed in these patents; instead, the evidence supports XNBR as a downstream elastomer platform whose performance is improved by nanocomposite processing and swellable reinforcement concepts.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20120035309A1
- Patent Title: Method to disperse nanoparticles into elastomer and articles produced therefrom
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a nanocomposite preparation method in which nanoparticles are first dispersed in a liquid form additive to form a nanoparticle pre-blend, and the pre-blend is then blended with an elastomer. The method is designed to improve nanoparticle dispersion in elastomers and can optionally use milling, high power probe sonication, or high shear homogenization to form the pre-blend. - Nanoparticle concentration: about 0.01 wt % to about 20 wt % — Detailed Description - Nanoparticle concentration: about 0.1 wt % to about 5 wt % — Detailed Description - Elastomer dissolution in solvent: greater than 90 wt % under the processing conditions — Detailed Description

Relevant Experimental Evidence - Example 1: Demonstrates the concept of dispersing nanoparticles in a liquid additive before blending with elastomer to improve dispersion. - Example 2: Identifies oils, plasticizers, and solvents as suitable liquid form additives for nanoparticle pre-blends. - Example 3: Shows that XNBR is among the elastomers explicitly contemplated for nanocomposite formation. - Example 4: Indicates that pre-blend formation can use milling, high power probe sonication, or high shear homogenization. - Example 5: Suggests solvent removal prior to blending and optional coagulation-based processing when elastomer is dissolved in a second solvent.

Technical Relevance This patent is directly relevant because it explicitly lists XNBR as a compatible elastomer and provides a practical formulation route for improving filler dispersion in XNBR-based compounds. It is especially useful for XNBR nanocomposite and downhole seal development where dispersion quality and barrier performance are critical.

Patent 2

- Patent Number: US8490707B2
- Patent Title: Oilfield apparatus and method comprising swellable elastomers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a swellable sealing system for subterranean wellbores in which a base polymer is combined with one or more reactive fillers, including a metal oxide, to form a seal assembly. Upon contact with a triggering

fluid, the assembly increases in volume and becomes less compliant; the disclosure also emphasizes that the swollen state may be permanent or non-deswelling. - Second volume retention: does not decrease to the first volume after termination of contact with triggering fluid — Claim 2 - Modulus increase: by a factor of one or more — Claim 4

Relevant Experimental Evidence - Example 1: Illustrates a sealing system that swells and stiffens in response to a triggering fluid. - Example 2: Shows that the swollen state can be retained after the triggering fluid is removed. - Example 3: Indicates that the rate of swelling can be controlled by reactive filler type, particle size, and concentration. - Example 4: Describes the use of a reinforcing reactive filler, including metal oxide, in the seal assembly. - Example 5: Connects the disclosure to downhole packers and other borehole sealing elements.

Technical Relevance This patent is relevant to XNBR because it addresses the same oilfield sealing application space in which XNBR is commonly used. The reactive-filler reinforcement concept is useful for designing XNBR-based swellable seals with improved stiffness, durability, and retention of sealing performance after swelling.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents are application-oriented rather than direct XNBR synthesis patents; they focus on downstream elastomer performance in harsh oilfield environments.
- US20120035309A1 centers on nanoparticle dispersion into elastomers using oils, plasticizers, or solvents, while US8490707B2 centers on reactive filler reinforcement and swelling behavior in seal assemblies.
- US20120035309A1 explicitly includes XNBR among the elastomers, whereas US8490707B2 discusses base polymers generically and emphasizes swellable packer materials rather than naming XNBR in the extracted evidence.
- US20120035309A1 discloses nanoparticle loading ranges of about 0.01 wt % to about 20 wt % and a preferred range of about 0.1 wt % to about 5 wt %, while US8490707B2 emphasizes filler type, particle size, and concentration as controls on swelling rate rather than giving a comparable composition range in the extracted text.
- US20120035309A1 uses pre-blend formation methods such as milling, probe sonication, and high shear homogenization; US8490707B2 uses reactive fillers, including metal oxides, to drive swelling and modulus increase.
- Both patents are aligned with downhole sealing, but US20120035309A1 is more relevant to XNBR compounding and nanocomposite processing, whereas US8490707B2 is more relevant to swellable seal design and post-swell mechanical reinforcement.

4. CONCLUSION

The evidence set does not disclose a direct XNBR polymerization or carboxylation synthesis route; instead, it identifies two important downstream technology directions for XNBR: nanoparticle-dispersed elastomer nanocomposites and reactive-filler-reinforced swellable seals. For XNBR compounding, the strongest disclosed processing guidance is to pre-disperse nanoparticles in a liquid additive such as oil, plasticizer, or solvent, then blend into the elastomer, with pre-blend formation by milling, probe sonication, or high shear homogenization. For swellable XNBR-based seals, the evidence supports using reactive fillers, including metal oxides, to increase volume and modulus after fluid exposure and to reduce deswelling. Recommended synthesis/compounding parameters based on the evidence are: use liquid pre-blend dispersion for fillers, target nanoparticle loadings in the disclosed 0.1-5 wt % preferred range when nanocomposite performance is desired, and consider reactive filler type, particle size, and concentration as the main levers for controlling swelling kinetics and stiffness in oilfield seal formulations.

5. REFERENCES

- US20120035309A1 | Method to disperse nanoparticles into elastomer and articles produced therefrom | Not disclosed | US | 2012 | <https://patents.google.com/patent/US20120035309A1/en>
- US8490707B2 | Oilfield apparatus and method comprising swellable elastomers | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8490707B2/en>